Farhan Akhtar Makandar

AI/ML Engineer | Deep Learning | Agentic AI | RAG Systems
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LinkedIn | GitHub | Hugging Face | Portfolio

Professional Summary

Final-year B.E. student in AI & Machine Learning with research internship experience at NIT Calicut. Specialized in transformer architectures, time-series forecasting, Agentic AI, and Retrieval-Augmented Generation (RAG). Developed a Temporal Fusion Transformer that achieved 22% higher predictive accuracy than XGBoost and KNN baselines for seismic response prediction. Proficient in PyTorch, LLM applications, multi-agent workflows, benchmarking, and production-oriented AI system development.

Experience

Research Intern – AI/ML Jun 2024 – Dec 2024
National Institute of Technology (NIT) Calicut

- Contributed to cutting-edge research on neural network optimization and deep learning for engineering systems.
- Designed and benchmarked transformer-based architectures (TFT, SAINT) using PyTorch; delivered a model that outperformed classical baselines by 22%.
- Built end-to-end training pipelines with feature engineering, hyperparameter tuning, and rigorous cross-validation.
- Conducted performance analysis, model validation, and prepared technical reports for internal review.
- Collaborated with a multidisciplinary team to integrate scientific computing techniques into predictive modeling.

Projects

Seismic Response Prediction using Temporal Fusion Transformer
PyTorch, Transformers, Time-Series Forecasting

- Built a Temporal Fusion Transformer (TFT) to predict seismic response in BRBF structures.
- Achieved 22% higher predictive accuracy over XGBoost and KNN baselines through systematic benchmarking and hyperparameter optimization.
- Developed an automated feature engineering pipeline and rigorous validation scheme, ensuring reproducibility.

AgentOS Multi-Agent AI Platform
Agentic AI, Multi-Agent Systems, Tool Calling

- Implemented multi-agent workflows with task decomposition, intelligent routing, tool invocation, and response synthesis.
- Designed agent communication protocols and reasoning pipelines that enable coordinated execution of complex tasks.
- Built a scalable orchestration engine, allowing specialized agents to collaborate efficiently on heterogeneous workloads.

RAG Pipeline with LLM Reasoning
RAG, Vector Search, LLM Applications

- Built an end-to-end Retrieval-Augmented Generation pipeline integrating embeddings, vector retrieval, and LLM reasoning.
- Implemented semantic search and context-aware response generation, significantly improving relevance and document understanding.
- Optimized retrieval strategies and prompt templates to reduce hallucination and enhance factual consistency.

Technical Skills

Languages: Python, Julia, R.
AI/ML: PyTorch, TensorFlow, Scikit-Learn, XGBoost, Transformers, TFT, SAINT
Generative AI: LLMs, RAG, Prompt Engineering, Multi-Agent Systems.
Research: Benchmarking, Experiment Tracking, Model Validation, Time-Series Forecasting

Tools: AWS, Hugging Face, Git, GitHub, Jupyter

Education

Yenepoya Institute of Technology, Moodbidri (VTU) 2022 – 2026 (Expected June 2026)
Bachelor of Engineering (B.E.) in Artificial Intelligence & Machine Learning

Certifications

- Oracle Cloud Infrastructure 2025 AI Foundations Associate
- NVIDIA DLI – Building RAG Agents with Large Language Models
- AWS Generative AI Essentials
- IBM Machine Learning with Python
- IBM Deep Learning Fundamentals